

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1065

Hard Rock 138 kV
December 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1065. The Interconnection Customer proposes a 82.4 MW solar photovoltaic facility interconnecting at the Hard Rock 138 kV substation in the Virginia area. The request was received on September 17, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 393 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$759,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Yarnen 345/138kV XFMR: 103.1% loading (321 MVA) following the outage of the McCartney Substation / Yarnen 345.0 kV Line; DFAX = 7.3%.

For the loss of Yarnen / McCartney 345.0 kV, the Sanger / E Longmeadow 138.0 kV Line loads to 100.9% of its 488 MVA rating. Project DFAX: 26.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yarnen 345/138kV XFMR	Replace transformer with higher-rated unit	\$15,967,000	3% (\$530,000)
East Longmeadow / Snger 138.0 kV	Reconductor line	\$2,258,000	10% (\$229,000)
	Total allocated to project		\$759,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.